

Moiré-induced altermagnetism from nonmagnetic constituents

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We propose a mechanism for nonmagnetic materials to develop altermagnetic order by moiré interference with nonmagnetic substrate, which is driven by structural relaxation and spontaneous twirls in moiré domain walls of lattice-mismatched moiré square lattices. When doped with one electron per moiré domain, a correlated insulating gap is opened by electron interaction. Depending on the location of the moiré potential minima, the moiré bands can show d -wave or g -wave altermagnetic splitting. The former can be enhanced by a finite twist angle; the latter is sensitive to strains that drive a transition to d -wave.

Introduction Altermagnetism is a recently identified class of compensated magnetic states in which opposite spin sectors are related by rotation and/or reflection [1–3], permitting momentum-dependent spin splitting of electronic bands with symmetry-protected touching points or nodal lines. Such types of band structure enable a range of exotic features, including spin-polarized currents [4, 5] and anomalous Hall effects [6–9] under zero net magnetization, which are promising ingredients for integrated electrical control of stray-field-free magnetic memory [4, 10, 11]; besides, both metallic and insulating altermagnets can display piezomagnetism without spin-orbit coupling (SOC) [12–15]. The richness of altermagnetic phenomena prompted an active search for material realizations [16–22]. However, crystalline atomic materials offer limited flexibility and tunability across the broad landscape of correlated physics. This calls for complementary routes to engineer altermagnetism in top-down designed artificial materials.

Moiré materials have been emerging as versatile platforms for engineering exotic phases of matter thanks to their exceptional tunability [23]. In van der Waals bilayers, a small twist angle or lattice mismatch produces a long-period moiré pattern that flattens the electronic bands and enhances electron interaction effects [24]. The resulting states include correlated insulators [25], superconductivity [26–28], Wigner crystals [29–31], and integer [32, 33] and fractional quantum anomalous Hall states [34–38]. The advancement of moiré engineering has motivated several designs of tunable altermagnetism via van der Waals coupling of *magnetic* constituents [39–49]. Given that both ferromagnetism [50] and antiferromagnetism [51, 52] have been realized with moiré stacking of *nonmagnetic* materials, a natural question is whether altermagnetism can be engineered in similar ways.

In this work, we report a minimal model study of moiré material consisting of *nonmagnetic* layers, which predicts emergent altermagnetism. Importantly, we engineer the altermagnetic symmetry by utilizing not only the electronic, but also the structural versatility of moiré materials – our work builds on discovery of $\sqrt{2} \times \sqrt{2}$ order in spontaneous twirls [53] in lattice-relaxed domain walls of

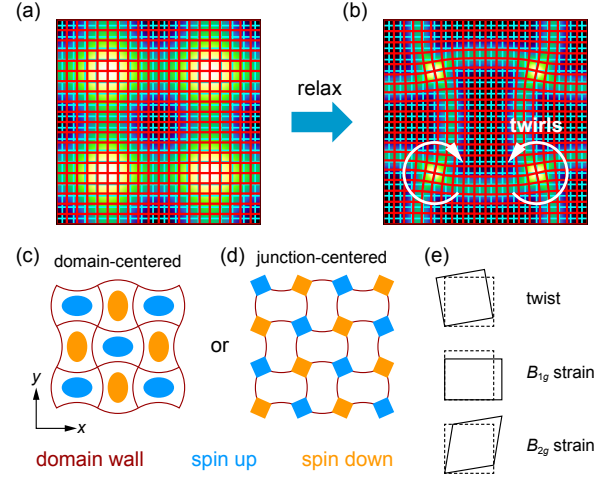


FIG. 1. (a), (b) Schematic of lattice relaxation of lattice-mismatched moiré square lattice from (a) the rigid configuration to (b) the stable configuration. The red and cyan grids respectively represent an elastic material layer and a rigid substrate, the former having a smaller natural lattice constant than the latter. In the background, darker (brighter) color represents regions with lower (higher) local interlayer coupling energy, or the moiré domain (domain wall) regions. The curved arrows mark the twirl chiralities. (c), (d) Schematics of Wannier orbitals of electron moiré bands in the two scenarios described on the top of each panel. The coordinate frame is marked at the bottom left of (c). The dark red lines represent the domain walls, and the blue and orange shapes represent occupied electron orbitals with different spin species, as illustrated on the bottom. (e) Schematics of twist and two types of strains applied to the elastic layer we consider in our work.

square-lattice moiré systems, demonstrating that emergent structural orders in moiré materials can give rise to novel electron orders.

Summary of mechanism As Figs. 1 (a), (b) show, moiré lattice relaxation expands regions with energetically favorable local interlayer stacking registry to domains, and shrinks other regions into domain walls [54–57]. The domain wall junctions further save energy by twirling, a phenomenon that can also occur in tri-

angular moiré domain structures [53, 58–64]. In the lattice-mismatched square lattice case relevant to our work, neighboring twirls tend to have opposite chirality, forming a staggered pattern in the square lattice (Fig. 1(b)). When the relaxed material is a semiconductor or point-node semimetal, the periodic moiré potential splits the electron bands into moiré bands [24, 65], whose Wannier centers depend on the locations of moiré potential minima. Two cases are schematically shown in Figs. 1(c), (d). Electron interaction in the bands gives rise to (extended) Hubbard model physics [66, 67] that generates both antiferromagnetic superexchange and ferromagnetic direct exchange couplings between electron spins of neighboring orbitals [68–70]. When the antiferromagnetic coupling dominates, because of the symmetry breaking by twirls, the system shows altermagnetic symmetry rather than conventional antiferromagnetic symmetry [71–73]. Assuming collinear spins and absence of SOC, we identify two cases: i) *domain-centered orbitals* (Fig. 1(c)), where the system has symmetries $[C_{2\perp}||C_{4z}]$, $[E||M_{\hat{x}}]$ and $[E||M_{\hat{y}}]$, realizing the spin Laue group $^{24}/^1m^2m^1m$ (notation adopted from Fig. 2 of Ref. [1]) with $d_{x^2-y^2}$ -wave spin splitting of moiré bands; ii) *junction-centered orbitals* (Fig. 1(d)), where the system has symmetries $[E||C_{4z}]$, $[C_{2\perp}||M_{\hat{x}}]$ and $[C_{2\perp}||M_{\hat{y}}]$, realizing the spin Laue group $^{14}/^1m^2m^2m$ with $g_{xy(x^2-y^2)}$ -wave spin splitting of moiré bands. Here we have adopted the standard convention for the symmetry operation notation $[A||B]$, where the spin (orbital) operation is on the left (right) [1]. $C_{2\perp}$ denotes spin flip, E denotes identity in spin sector, C_{4z} denotes 4-fold in-plane counterclockwise spatial rotation of orbitals about a domain junction point, and $M_{\mathbf{n}}$ is spatial reflection of orbitals about a mirror that passes a domain center and is normal to vector $\mathbf{n}$. In both cases, the system also has the symmetry of spin-flipping diagonal reflection combined with moiré lattice translation: $[C_{2\perp}||M_{\hat{x}+\hat{y}}|\tau_m]$ and $[C_{2\perp}||M_{\hat{x}-\hat{y}}|\tau_m]$, which are products of the four-fold rotation and normal-mirror reflection symmetries. τ_m is the primitive translation of the pre-relaxation moiré, *i.e.*, half diagonal translation of the relaxed $\sqrt{2} \times \sqrt{2}$ superlattice.

We will show in the rest of this paper that when the effective electron mass of the parent semiconductor band is relatively small, the d -wave spin splitting in the domain-centered case can reach experimentally detectable range, whereas the g -wave spin splitting in the junction-centered case is expected to be rather weak. Before elaborating on our model settings, we note that the altermagnetic order is tunable with twist angle and strain engineering. Schematics of twist angle and strains are shown in Fig. 1(e). In the domain-centered case, a finite twist angle between the mismatched layers breaks all the mirror symmetries but preserves $[C_{2\perp}||C_{4z}]$, reducing the $^{24}/^1m^2m^1m$ spin Laue group to $^{24}/^1m$ with the d -wave spin splitting that is a mixture of $d_{x^2-y^2}$ - and d_{xy} -wave components. We will see that such a twist can enhance

altermagnetic band splitting and related properties. In the junction-centered case, a B_{1g} (B_{2g}) strain exerted on one layer breaks $[E||C_{4z}]$ and $[C_{2\perp}||M_{\hat{x}\pm\hat{y}}|\tau_m]$ ($[E||C_{4z}]$, $[C_{2\perp}||M_{\hat{x}}]$ and $[C_{2\perp}||M_{\hat{y}}]$) while preserving the remaining spin-flip reflection symmetries. The $^{14}/^1m^2m^2m$ spin Laue group hence reduces to $^2m^2m^1m$ with d_{xy} - or $d_{x^2-y^2}$ -wave band splitting in the B_{1g} or B_{2g} case, respectively. In the End Matter, we will see that for both types of strains the d -wave order begins to dominate over the original g -wave order from very tiny strain strengths.

Model. Our model consists of lattice relaxation part and electron part. For lattice, we use the continuum elasticity formulation [53, 57, 74, 75] of an elastic layer on rigid substrate, which minimizes the sum of elastic and substrate coupling energies by varying the in-plane local displacement function $\mathbf{u}(\mathbf{r}) = (u_x(\mathbf{r}), u_y(\mathbf{r}))$ under moiré super-periodic constraints defined by the lattice constant mismatch ϵ , the twist angle θ , and the B_{1g} /orthorhombic and B_{2g} /shear strains (see Fig. 1(e)) exerted to the elastic layer prior to relaxation. Throughout this work we let the natural lattice constant of the elastic material be $\epsilon = 5\%$ larger than the substrate and allow θ , B_{1g} and B_{2g} to vary. Systems with other values of ϵ , including negative values, are related via rescaling of model parameters [76].

We characterize the stiffness of the elastic material with three modulus coefficients λ , μ and μ_s , the former two being the Lamé coefficients, and μ_s an independent component of shear modulus allowed by the symmetry of square lattices. See Supplemental Material [76] for detailed definitions. Ratios between λ , μ and μ_s vary widely across materials [77–80]. In major part of this work we take the isotropic limit $\mu_s = \mu$ and the 2D Cauchy relation limit $\lambda = \mu$. The coupling strength to the substrate is characterized by a single real parameter V_1 , which is the first-star Fourier coefficient of the coupling energy density $V(\mathbf{d})$ [53, 76] as a function of the local stacking registry $\mathbf{d} = \mathbf{d}(\mathbf{r}) = \mathcal{M}\mathbf{r} + \mathbf{u}(\mathbf{r})$. Here $\mathcal{M}\mathbf{r}$ is the spatially linear pre-relaxation local stacking registry caused by ϵ , θ , B_{1g} and/or B_{2g} . Domain wall twirling only happens when V_1/μ is above a threshold that is at the order of ϵ^2 [53], and the $\sqrt{2} \times \sqrt{2}$ order in twirls illustrated in Fig. 1(b) is only stable when V_1/μ is below a larger threshold, above which further breaking of periodicity occurs.

For electron modeling, we assume the general spin-degenerate semiconductor form of electrons or holes in the elastic material, $H_K = \mathbf{p}^2/2m$, acting on the envelope functions in a single time-reversal-invariant valley with no other flavors than spin. This description can be appropriate not only for SOC-free systems, but also for centrosymmetric nonmagnetic materials with SOC, where the spin degeneracy is protected by inversion and time-reversal symmetries. Here m is the effective mass, ranging from $0.01m_e$ to several times m_e in real materials [77, 81] where m_e is the free electron mass. A typical moiré kinetic energy scale is $E_m = \epsilon^2\hbar^2/2ma^2$, which is

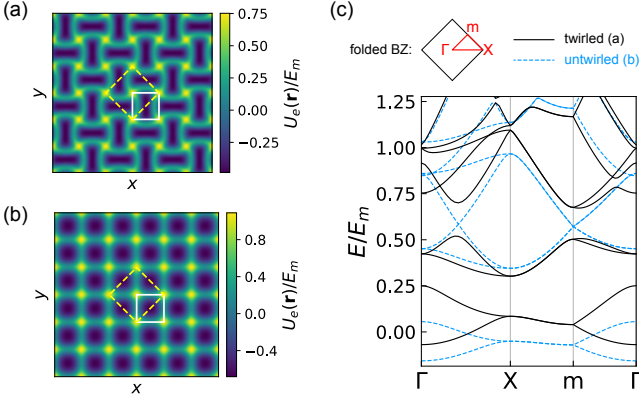


FIG. 2. (a), (b) The electron potential $U_e(\mathbf{r}) = D\nabla \cdot \mathbf{u}(\mathbf{r})$ obtained under $\epsilon = 0.05$, $\theta = B_{1g} = B_{2g} = 0$, $V_1/\mu = 0.004$ and $D = 10E_m$ via lattice relaxation performed with (a) $\sqrt{2} \times \sqrt{2}$ supercell and (b) the original moiré periodicity. The dashed (solid) square is the doubled (original) moiré cell. (c) The non-interacting moiré band structure along the red high-symmetry lines shown on the top. The black solid and blue dashed lines respectively represent the band structure under the electron potential shown in (a) and (b). The bands are spin-degenerate in the non-interacting case. The total four-fold degeneracy along Xm is protected by the non-symmorphic and time reversal symmetries of the non-interacting system, and reduces to two-fold spin degeneracy when the magnetism breaks the time reversal symmetry.

about 235meV for $m = 0.1m_e$, $a = 4\text{\AA}$ and $\epsilon = 0.05$. Relaxed moiré pattern generally affects electrons via deformation potential $D\nabla \cdot \mathbf{u}(\mathbf{r})$, substrate potential $\Delta(\mathbf{d}(\mathbf{r}))$ and pseudo-magnetic field $\mathbf{p} \cdot \mathbf{A}(\mathbf{r})/2m + H.c.$ [66, 82–84]. Pseudo-magnetic field is absent in time-reversal-invariant valleys; in the first-star approximation, the substrate and deformation potentials have similar shapes hence can be effectively merged for our purpose. Therefore, we let the non-interacting part of our effective Hamiltonian be $H_0 = H_K + D\nabla \cdot \mathbf{u}$, where the effect of $\Delta(\mathbf{d}(\mathbf{r}))$ is absorbed into the deformation potential D . D can be either positive or negative [85], which respectively corresponds to the domain- or junction-centered case in our work. Fig. 2(a) shows an example of deformation potential $D\nabla \cdot \mathbf{u}(\mathbf{r})$ obtained under $D > 0$, while Fig. 2(b) shows its untwisted counterpart. The $\sqrt{2} \times \sqrt{2}$ periodicity in twirls allows us to obtain the moiré band structure in the folded moiré Brillouin zone (BZ). One non-interacting example is shown in Fig. 2(c), where we see that the twirls open a significant gap between the 4th and 5th moiré bands.

Next, we let the electrons interact via Coulomb potential $C(\mathbf{r}) = e^2/\tilde{\epsilon}|\mathbf{r}|$, where $\tilde{\epsilon}$ is the dielectric constant of the environment. A typical moiré Coulomb energy scale is $C_m = \epsilon e^2/\tilde{\epsilon}a$, which is about 36meV for $\tilde{\epsilon} = 5$, $a = 4\text{\AA}$ and $\epsilon = 0.05$. The spin-resolved mean-field band structures are obtained via self-consistent Hartree-Fock theory on a discretized BZ grid containing $12\sqrt{2} \times 12\sqrt{2} = 288$

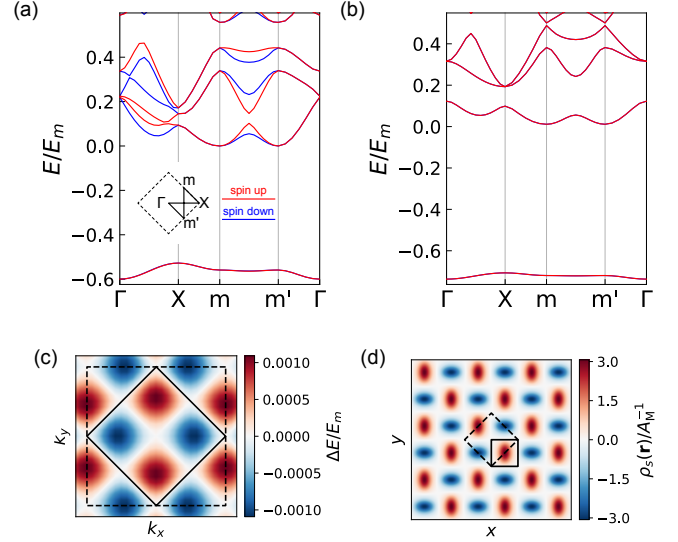


FIG. 3. (a) The spin-resolved mean-field moiré band structure under $\epsilon = 0.05$, $\theta = B_{1g} = B_{2g} = 0$, $V_1/\mu = 0.004$, $D = 10E_m$, interaction strength $C_m = 0.2E_m$ and filling factor corresponding to one electron per $\sqrt{2} \times \sqrt{2}$ supercell per spin sector (i.e., one filled band in Fig. 2(c) for each spin), plotted along the high-symmetry line shown in the inset. (b) Same as (a), but under the original-moiré-periodicity relaxation background illustrated in Fig. 2(b). Here the bands are spin-degenerate. (c) The spin splitting $\Delta E_k = E_k^\uparrow - E_k^\downarrow$ of the lowest band in (a). The solid (dashed) black square is the BZ corresponding to the supercell (the original moiré cell). (d) The spin density, $\rho_s(\mathbf{r}) = \rho_\uparrow(\mathbf{r}) - \rho_\downarrow(\mathbf{r})$, of the lowest band in (a) in units of one per original unit cell area $A_M = a^2/\epsilon^2$.

sample points. Detailed formulation is presented in the Supplemental Material [76]. Computer code for our computation is generated by OpenAI Codex with GPT 5.5 model under human-written instruction that contains all mathematical details.

Interacting results. We present our mean-field results in the $D > 0$ case in Fig. 3. At band filling corresponding to one electron per $\sqrt{2} \times \sqrt{2}$ supercell in each spin sector, the Coulomb repulsion opens a large correlated insulating gap between the lowest two bands in Fig. 2(c). Fig. 3(a) shows spin splitting up to $0.05 \sim 0.1E_m$ with nodal lines along $X - m$ and $m' - \Gamma$, consistent with $d_{x^2-y^2}$ -wave symmetry. The spin splitting is caused by the twirls in the relaxed domain walls, because Fig. 3(b) shows that the mean-field moiré bands are spin-degenerate under the twirl-free relaxation background. In Fig. 3(a), the lowest band has small spin splitting, which is shown in Fig. 3(c). The spin density map of the filled band presented in (d) confirms the signature of the schematic lattice model in Fig. 1(c). We also show in the Supplemental Material [76] that d -wave spin splitting occurs not only at this particular parameter combination, but also in a range of parameters.

As a finite twist angle between the elastic material

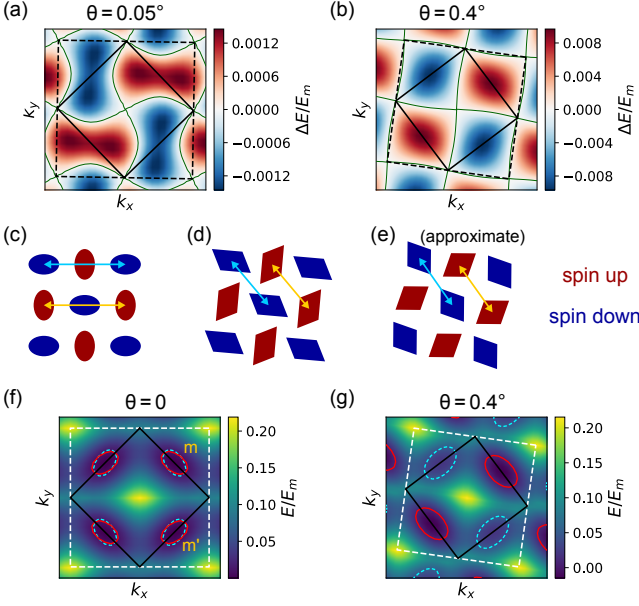


FIG. 4. (a), (b) The spin splitting of the lowest mean-field band of a system with the same parameters as the one shown in Fig. 3(c), except for a finite twist angle between the elastic material and the substrate illustrated at the top of each panel. Here the curved lines outline the band touching. (c)-(e) Schematics of effective lattice models that describe the lowest bands under (c) $\theta = 0$, (d) relatively small finite θ , and (e) relatively large θ , where the arrows illustrate the nearest spin-inequivalent hopping in the three cases. The shapes are only schematic and reflect the symmetries of the system. The mirror symmetries of (e) are approximate in the physical systems it points to. (f), (g) The dispersion of the first spin-up moiré band above the correlated insulating gap, in a system with twist angle illustrated on the top of each panel and the rest of the parameters identical to (a), (b). The red solid (cyan dashed) ellipses are the spin-up (spin-down) Fermi surfaces upon doping extra 0.1 electrons per original moiré cell, ignoring residual interaction effects of the doping electrons.

and substrate is turned on, the spin nodal lines become curved due to violation of mirror symmetry, and ultimately rearranges the spin splitting of the lowest band to nearly d_{xy} -wave (Figs. 4 (a), (b)). In addition, the spin splitting of the lowest band is enhanced by the twist angle by nearly an order of magnitude. Since the twist angle breaks mirror symmetries but preserves C_{4z} , the spin degeneracy at Γ is maintained; Along any loop around Γ , the spin splitting has to alternate 4 times between positive and negative, guaranteeing the presence of nodal lines.

Figs. 4 (c)-(e) present schematic understanding of the twist angle evolution of band splitting. At zero twist, the nearest same-spin-sublattice neighbor hopping has to take a universal value for both spin sublattices as enforced by $[C_{2\perp}||C_{4z}]$ and mirror symmetries, hence the nearest spin-inequivalent hopping is the second same-sublattice neighbor hopping, as illustrated by the arrows

in Fig. 4(c). For the lowest band with s -orbital-like Wannier orbitals, the hopping amplitudes decay exponentially with distance, hence the effect of spin contrast in this hopping is exponentially suppressed, giving almost degenerate band structures. At finite twist angle, violation of mirror symmetries allows the nearest same-sublattice neighbors to have spin-inequivalent hoppings, as Fig. 4(d) shows. As the twist angle continues to increase, the spin contrast in the nearest same-sublattice hopping amplitudes grows up to orders of magnitude above the spin contrast in further-neighbor hoppings, thus dominates the spin splitting in the band structure. Ignoring the spin contrast in further-neighbor hoppings, the system has approximate spin-flipping mirror symmetries $[C_{2\perp}||M_{\hat{x}}]$ and $[C_{2\perp}||M_{\hat{y}}]$ (Fig. 4(e)), leading to approximate d_{xy} splitting. In the junction-centered case ($D < 0$), a similar mechanism of spin-splitting enhancement and near-neighbor dominance, driven by strain-induced symmetry lowering and leading to g -to- d -wave transition [86–88], is described in End Matter.

One important experimental signature of d -wave altermagnetic metals is anisotropic transport properties with spin sectors related via 90° rotation [5, 71]. In our domain-centered system, we consider further doping our correlated insulating states with electrons to enable electron conduction. To roughly assess the spin- 90° -related anisotropy effects, we assume that the added electrons do not interact but simply form Fermi surfaces in the mean-field moiré bands of the correlated insulator. Fig. 4(f) shows slightly mismatching Fermi surfaces for spin-up and spin-down. With a finite twist angle, the energetic symmetry between m and m' mini-valleys in a spin-resolved band is lifted, allowing mini-valley-polarized population in each spin sector (Fig. 4(g)). The anisotropy of elliptic Fermi surfaces manifest as spin-contrasting anisotropy in effective band mass, suggesting spin-related 90° anisotropic transport. Though we do not address them explicitly, we note that correlations can induce additional exotic behaviors including unconventional superconductivity, charge density wave and strange metal on top of altermagnetism [89, 90]. These phases may cooperate or compete with altermagnetism to create even more interesting phase diagrams.

Discussion. We have demonstrated path to realization of correlated insulating altermagnet in moiré materials consisting of nonmagnetic layers, within a minimal electronic continuum model setting built upon continuum elasticity model of lattice relaxation. The altermagnetic spin-splitting originates from the interplay of antiferromagnetic superexchange in electrons and staggered twirls in square lattice moiré domain walls. According to our results, when the typical moiré kinetic energy scale $E_m = \epsilon^2 \hbar^2 / 2ma^2$ is around 0.1 times the deformation potential constant D and the typical moiré Coulomb energy scale $C_m = \epsilon \epsilon^2 / \bar{\epsilon} a$ is $\sim 0.2 E_m$, the empty mean-field bands have altermagnetic spin split-

TABLE I. First-principle-based effective band masses and deformation potential constants of monolayer square-lattice M_2X from Ref. [77]. Note a factor 2 between their and our conventions of the deformation potential constant, and that the sign of D for valence bands is flipped relative to Ref. [77].

material	band	m/m_e	$E_1 = 2D$ (eV)
Ag ₂ S	conduction	0.18	2.35
Ag ₂ Se	conduction	0.19	1.97
Au ₂ S	conduction	0.08	1.95
	valence	0.12	4.33
Au ₂ Se	conduction	0.08	1.68
	valence	0.14	3.11

ting $\sim 0.05E_m$. In real materials $D \sim 1$ eV [77, 85], meaning that when $E_m \sim 100$ meV and $C_m \sim 20$ meV (i.e. when $(\sqrt{m/m_e})(a/\epsilon) \sim 40$ Å and $\tilde{\epsilon}a/\epsilon \sim 720$ Å), the altermagnetic spin splitting is ~ 5 meV, which is detectable with high-resolution spin-polarized scanning tunneling microscopy (STM) [91, 92]. By contrast, since the structural energy saved by twirling is much larger, ~ 1 eV [53], the altermagnetic splitting in electron bands is not expected to significantly affect the moiré lattice structure.

Taking the experimentally realistic dielectric constant $\tilde{\epsilon} \sim 5$, the parameter regime specified above suggests a moiré length scale $a/\epsilon \sim 144$ Å and an effective mass $m \sim 0.08m_e$. Hence, for actual material realizations we target our search at 2D nonmagnetic centrosymmetric square-lattice semiconductors with light electrons or holes in a single valley. Plausible candidates include square-lattice M_2X where $M = \text{Ag, Au}$ and $X = \text{S, Se}$ [77], whose parameters are summarized in Table I. Additional requirements include: (i) the substrate needs to be an insulator whose bands are out of the way of the altermagnetic moiré bands; (ii) the lattice mismatch ϵ and the structural coupling strength V_1 are suitable so that the $\sqrt{2} \times \sqrt{2}$ twirl pattern in the domain wall network is energetically stable. We leave in-depth investigation of actual material realization for future work.

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END MATTER

The junction-centered ($D < 0$) mean-field results are presented in Fig. 5. In the absence of twist or strain, the bands show g -wave splitting, with spin degeneracy along $m' - \Gamma - X - m$, as shown by Figs. 5 (a), (b). The band splitting is significantly smaller than in the $D > 0$ case presented in Fig. 3(a), and is as small as $\sim 10^{-7} E_m$ for the lowest band (Fig. 5(b)). Figs. 5 (c), (d) show that applying strain to the elastic layer can enhance the spin splitting, and that the enhancing effects of B_{1g} strain on the low bands are stronger than those of B_{2g} . Figs. 5 (e), (f) ((g), (h)) show the evolution of the lowest-band spin splitting with the onset of B_{2g} (B_{1g}) strain. We see that the spin splitting map is already significantly reconstructed under extremely weak ($10^{-5} \sim 10^{-7}$) strains. As the strain grows, the g -wave splitting ultimately transitions to $d_{x^2-y^2}$ -wave (d_{xy}) in the B_{2g} (B_{1g}) case, which is consistent with the symmetry analysis in the *Summary of mechanism* section of the main text.

Like the twist angle evolution in domain-centered systems, here the tininess and strain sensitiveness of g -wave splitting can also be understood with schematics of effective lattice presented in Figs. 5 (i)-(k). In the absence of strain, even the nearest spin-inequivalent hopping spans three moiré cells – all nearer-neighbor hoppings are spin-equivalent by symmetry, as Fig. 5(i) illustrates. Under a B_{2g} (B_{1g}) strain, the nearest spin-contrasting hopping becomes the second-neighbor (nearest-neighbor) same-sublattice hopping, as shown in Fig. 5(j) (Fig. 5(k)), which dominates exponentially over the original spin contrast that gives the g -wave band splitting. Moreover, that the nearest-neighbor spin contrast in hopping is closer under B_{1g} strain than under B_{2g} strain also explains why B_{1g} enhances low-band spin splitting more than B_{2g} does, and why the g -wave order is more vulnerable to B_{1g} strain than to B_{2g} strain.

Similarly to the twisted domain-centered system discussed in the main text (Fig. 4(g)), we show in Figs. 5 (l), (m) that a B_{1g} strain can induce significant spin contrast in Fermi surface and transport anisotropy while

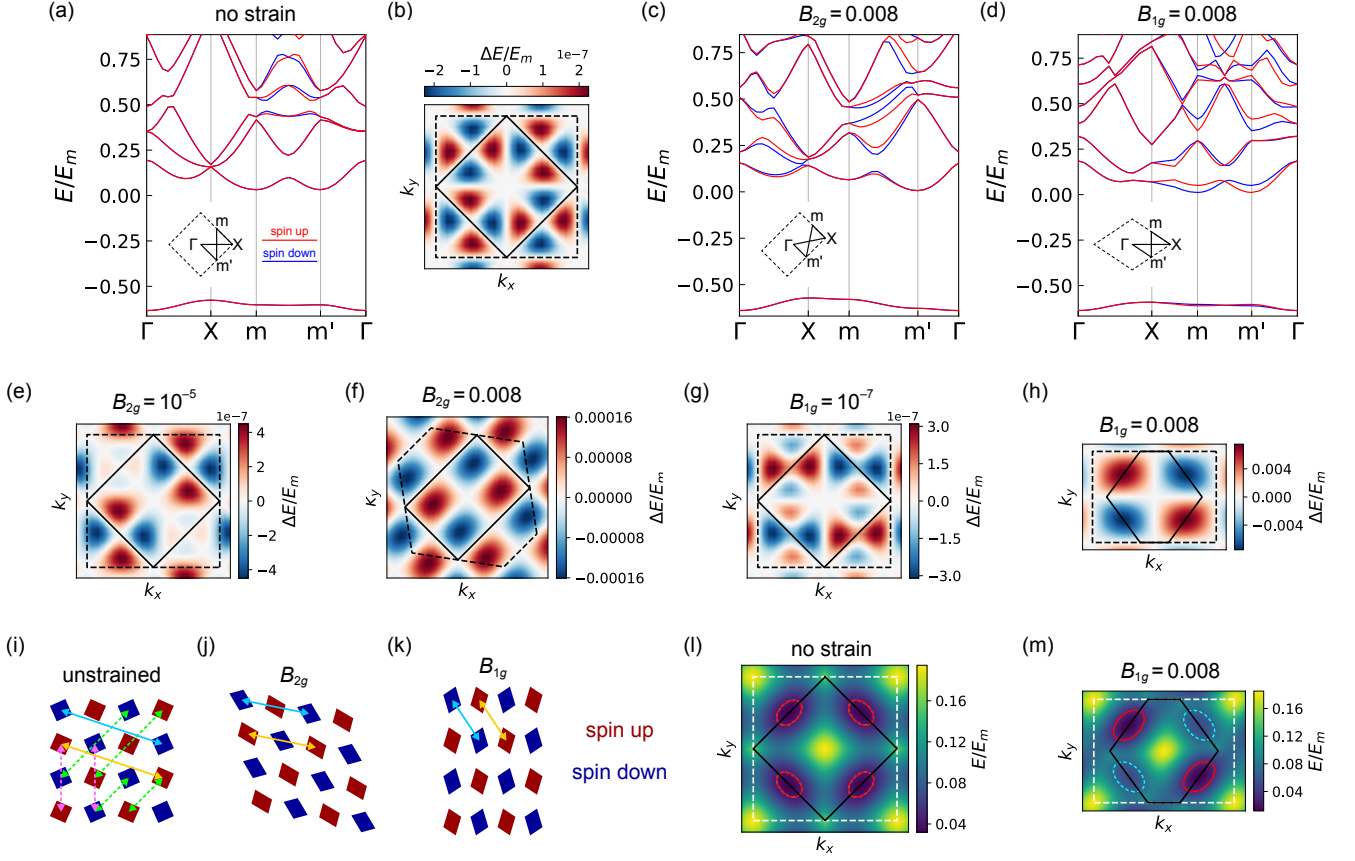


FIG. 5. (a) The spin-resolved mean-field moiré band structure of the system with the same parameters as in Fig. 3(a) except that here $D = -10E_m$ instead of $10E_m$. (b) The spin splitting of the lowest band in (a). The solid (dashed) black square is the first BZ of the $\sqrt{2} \times \sqrt{2}$ supercell (the original moiré cell). (c), (d) Same as (a) except for nonzero strains marked on the top. The dashed line in the inset of (d) is a direct affine transform from the unstrained moiré BZ, which is not the first BZ of the strained moiré lattice. However, the vertices of the plotting line are still time-reversal-invariant. (e)-(h) Same as (b) except for nonzero strains marked on the top. (i)-(k) Schematics of effective lattice models that describe the low bands with the strain condition indicated on the top. The yellow and blue solid arrows illustrate the nearest spin-inequivalent hopping in each case. The green and pink dashed arrows in (i) are pairs of spin-equivalent hoppings. The shapes are only schematic and reflect the symmetries of the system. (l), (m) The dispersion of the first spin-up moiré band above the correlated insulating gap, in a system with (l) no strain and (m) $B_{1g} = 0.008$ and the rest of the parameters identical to (a), (b). Again, the red solid (cyan dashed) ellipses are the spin-up (spin-down) Fermi surfaces upon doping extra 0.1 electrons per original moiré cell, ignoring residual interaction effects of the doping electrons. In (l), the Fermi surfaces are almost spin-identical because of the negligible g -wave splitting.

the correlated insulating state is further electron-doped. Note that due to violation of 4-fold rotation symmetry by strain, here the spin-resolved transport anisotropies may not be strictly related via 90° rotation.

SUPPLEMENTAL MATERIAL FOR “MOIRÉ-INDUCED ALTERMAGNETISM FROM NONMAGNETIC CONSTITUENTS”

Details of lattice relaxation modeling

The total lattice energy $E_l[\mathbf{u}] = \int d^2\mathbf{r}(U + V)$ is a functional of the local displacement $\mathbf{u}(\mathbf{r})$. Here

$$U = \frac{\lambda}{2}(\nabla \cdot \mathbf{u})^2 + \mu [(\partial_x u_x)^2 + (\partial_y u_y)^2] + \frac{\mu_s}{2}(\partial_x u_y + \partial_y u_x)^2 \quad (\text{S1})$$

is the elastic energy density, where λ and μ are the regular Lamé coefficients, and μ_s is the shear Lamé coefficient. A different set of notation, C_{11} , C_{12} and C_{66} , is often used in past work [77–79] and relates to our notation via

$$\lambda = C_{12}, \quad \mu = \frac{C_{11} - C_{12}}{2}, \quad \mu_s = C_{66}. \quad (\text{S2})$$

First-principle values for some materials are listed in Table II.

$$V = V(\mathbf{d}(\mathbf{r})) = \sum_{\mathbf{G}} V_{\mathbf{G}} e^{i\mathbf{G} \cdot \mathbf{d}(\mathbf{r})} \quad (\text{S3})$$

is the coupling energy density to the substrate, which is locally a periodic function of the stacking registry vector $\mathbf{d}$ with the periodicity of *strained* (if B_{1g} or B_{2g} is nonzero) pre-relaxation lattice of the elastic material. $V_{\mathbf{G}}$ is the Fourier coefficient of $V(\mathbf{d})$. Given a displacement configuration $\mathbf{u}(\mathbf{r})$, the local stacking configuration is $\mathbf{d}(\mathbf{r}) = \mathcal{M}\mathbf{r} + \mathbf{u}(\mathbf{r}) = (\mathcal{M}_0 + \mathcal{M}_B)\mathbf{r} + \mathbf{u}(\mathbf{r})$, with

$$\mathcal{M}_0 = \begin{pmatrix} \epsilon & -\theta \\ \theta & \epsilon \end{pmatrix}, \quad \mathcal{M}_B = \begin{pmatrix} B_{1g} & B_{2g} \\ B_{2g} & -B_{1g} \end{pmatrix}, \quad (\text{S4})$$

where ϵ is the lattice mismatch ratio, θ is the twist angle, and B_{1g} and B_{2g} are respectively the orthirhombic and diagonal area-preserving strains imposed on the elastic layer prior to relaxation. We note that the periodicity of $V(\mathbf{d})$ with respect to $\mathbf{d}$ is affected by B_{1g} and B_{2g} via affine deformation $1 + \mathcal{M}_B$, hence specify that $V(\mathbf{d}) = V_0((1 + \mathcal{M}_B)^{-1}\mathbf{d})$, and truncate the Fourier components of $V_0(\mathbf{d})$ to the first star of reciprocal lattice as in Ref. [53], capturing the substrate coupling strength with a single real parameter V_1 . In particular,

$$V_0(\mathbf{d}) = 2V_1 \left(\cos \frac{2\pi d_x}{a} + \cos \frac{2\pi d_y}{a} \right), \quad (\text{S5})$$

where a is the natural microscopic lattice constant of the elastic layer. The moiré cell is spanned by $\mathbf{A}_1 = \mathcal{M}^{-1}\mathbf{a}_1$ and $\mathbf{A}_2 = \mathcal{M}^{-1}\mathbf{a}_2$, where $\mathbf{a}_{1,2}$ span the strained microscopic cell, *i.e.*, $\mathbf{a}_j = (1 + \mathcal{M}_B)\mathbf{a}_j^0$ with $\mathbf{a}_1^0 = (a, 0)$, $\mathbf{a}_2^0 = (0, a)$.

TABLE II. Computationally predicted lattice constants and stiffness coefficients of some monolayer square-lattice materials. Here C_{11} , C_{12} , C_{66} and μ are in units of eV/Å².

material	reference	lattice constant (Å)	C_{11}	$C_{12} = \lambda$	$C_{66} = \mu_s$	$\mu = (C_{11} - C_{12})/2$	λ/μ	μ_s/μ
s(II)-Cu ₂ S	[77]	5.02	2.103	0.169	1.211	0.967	0.175	1.252
s(II)-Cu ₂ Se	[77]	4.97	2.303	0.474	1.080	0.914	0.519	1.182
s(I)-Ag ₂ S	[77]	5.88	1.454	0.193	0.668	0.63	0.306	1.06
s(I)-Ag ₂ Se	[77]	5.9	1.311	0.193	0.668	0.559	0.345	1.195
s(I)-Au ₂ S	[77]	5.81	1.972	0.63	0.949	0.671	0.939	1.414
s(I)-Au ₂ Se	[77]	5.82	1.373	0.499	0.868	0.437	1.142	1.986
s(I)-Au ₂ Te	[78]	5.85	1.464	0.372	0.768	0.546	0.681	1.407
s(II)-Au ₂ Te	[78]	5.61	2.172	1.186	0.886	0.493	2.406	1.797
monolayer FeSe	[79]	5.32	4.955	0.328	1.218	2.313	0.142	0.527

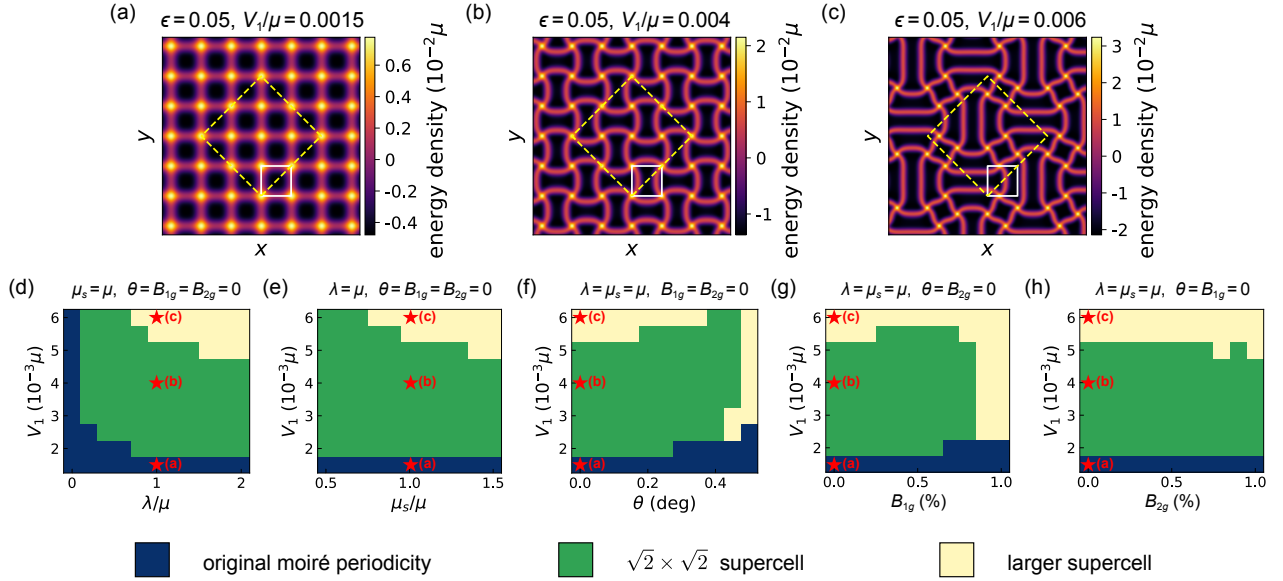


FIG. S1. (a)-(c) The local energy density configuration $U(\mathbf{r}) + V(\mathbf{d}(\mathbf{r}))$ of the lowest-energy solution of the relaxation equilibrium equation Eq. (S6), under $\lambda = \mu_s = \mu$, $\epsilon = 0.05$, $\theta = B_{1g} = B_{2g} = 0$ and V_1/μ indicated on the top of each panel. The solid square is the original moiré cell, and the dashed square is the $2\sqrt{2} \times 2\sqrt{2}$ supercell we allow in our relaxation calculation. (d)-(h) Phase diagram of spontaneous moiré-translational-symmetry breaking with respect to V_1 and various other parameters. In all of them, $\epsilon = 0.05$ and other fixed parameters are labeled on the top of each panel. The red stars mark the positions of the state indicated by (a)-(c) in the phase diagram.

According to Eqs. (S1) and (S3), the variational condition $\delta E_I / \delta \mathbf{u} = \mathbf{0}$ gives the nonlinear derivative equation

$$\mu \nabla^2 \mathbf{u} + (\lambda + \mu) \nabla (\nabla \cdot \mathbf{u}) + (\mu_s - \mu) (\partial_y, \partial_x) (\partial_x u_y + \partial_y u_x) = i \sum_{\mathbf{G}} \mathbf{G} V_{\mathbf{G}} e^{i\mathbf{G} \cdot (\mathcal{M}\mathbf{r} + \mathbf{u}(\mathbf{r}))}, \quad (\text{S6})$$

which is numerically solved with the iteration method described in Ref. [53] assisted with Anderson acceleration [93], allowing for moiré translational symmetry breaking up to $2\sqrt{2} \times 2\sqrt{2}$ moiré cells. Figs. S1 (a)-(c) show three typical equilibrium states of relaxation that have different periodicities. We see that the system tends to preserve its original moiré periodicity under weak V_1 , while stronger V_1 favors moiré translational symmetry breaking into larger cells. Figs. S1 (d)-(h) shows that this trend is general under a range of other model parameters including the Lamé coefficients, twist angle and strains.

Self-consistent Hartree-Fock theory

We perform Hartree-Fock calculation for electrons under Coulomb interaction and moiré potential with $\sqrt{2} \times \sqrt{2}$ supercell, assuming no spin mixing or further translational symmetry breaking by electron interaction. The spin-resolved moiré bands are governed by the spin-projected mean-field Bloch Hamiltonians, $H_{\text{MF}}^\sigma(\mathbf{k}) = H_0(\mathbf{k}) + \Sigma_{\text{H}} + \Sigma_{\text{F}}^\sigma(\mathbf{k})$, where $\sigma = \uparrow, \downarrow$ respectively stand for spin up and spin down, $\mathbf{k}$ is the moiré momentum, Σ_{H} and $\Sigma_{\text{F}}^\sigma(\mathbf{k})$ are respectively the Hartree and Fock self energy matrices while only the latter can be $\mathbf{k}$ - and spin-dependent. The non-interacting part of the Hamiltonian, $H_0(\mathbf{k})$, has matrix elements

$$H_{0,gg'}(\mathbf{k}) = \frac{\hbar^2(\mathbf{k} + \mathbf{g})^2}{2m} \delta_{gg'} + iD(\mathbf{g} - \mathbf{g}') \cdot \mathbf{u}_{\mathbf{g}}, \quad (\text{S7})$$

where $\mathbf{g}$ is a reciprocal lattice vector of the $\sqrt{2} \times \sqrt{2}$ superlattice, and $\mathbf{u}_{\mathbf{g}}$ is the corresponding Fourier coefficient of $\mathbf{u}(\mathbf{r})$. The self-energy matrices is associated with the spin-resolved one-particle density matrices $\rho^\sigma(\mathbf{k})$ by

$$\Sigma_{\text{H},gg'} = C_{\mathbf{g}-\mathbf{g}'} (\rho_{\mathbf{g}-\mathbf{g}'}^\uparrow + \rho_{\mathbf{g}-\mathbf{g}'}^\downarrow), \quad \rho_{\mathbf{g}}^\sigma = \frac{1}{(2\pi)^2} \sum_{g'} \int_{\text{mBZ}} d^2 \mathbf{q} \rho_{(\mathbf{g}+\mathbf{g}')g'}^\sigma(\mathbf{q}) \quad (\text{S8})$$

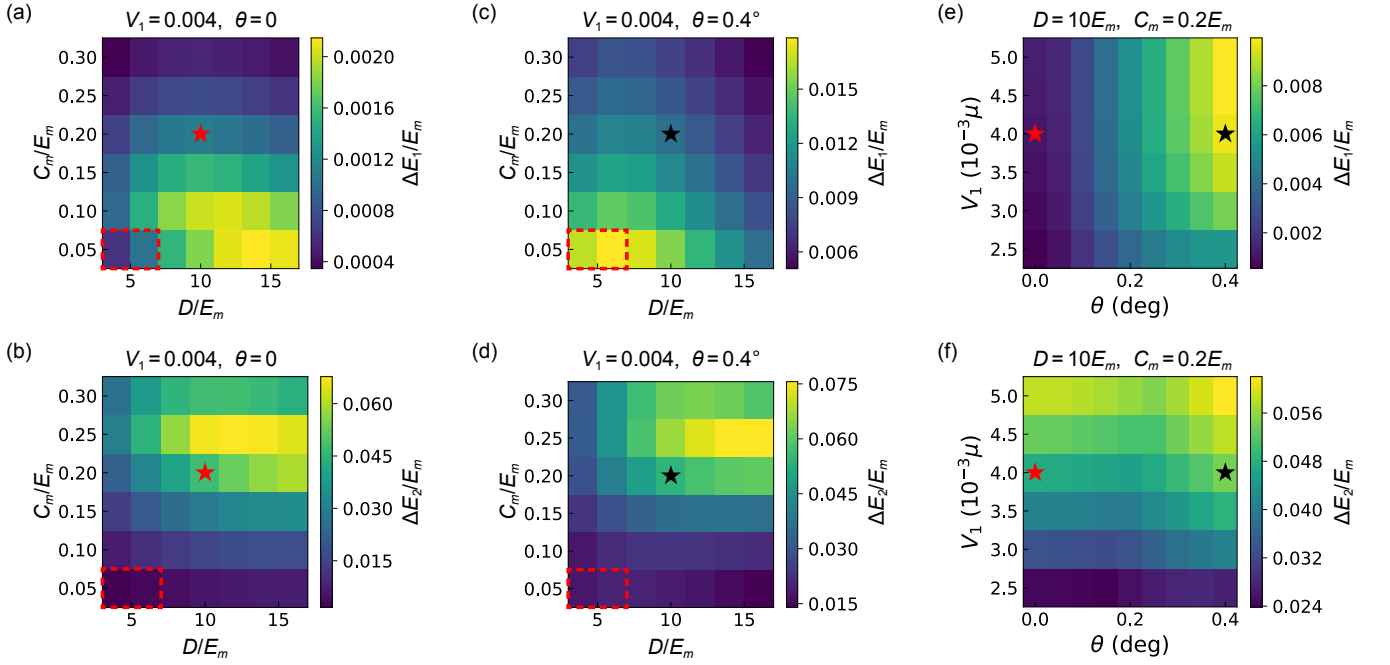


FIG. S2. Parameter dependence of maximum spin splitting $\Delta E_n = \max_{\mathbf{k}} |\Delta E_{n\mathbf{k}}|$ of (a)-(c) the filled moiré band ($n = 1$) and (d)-(f) the first empty moiré band ($n = 2$) in the domain-centered system, with fixed and sweeping parameters indicated on the top and axes labels of each panel. Here $\epsilon = 0.05$, $\lambda = \mu_s = \mu$ and $B_{1g} = B_{2g} = 0$. The red (black) stars mark the position of the parameter combination under which results are presented in main text Fig. 3 (Fig. 4(b)). The red dashed boxes mark the regions where the Hartree-Fock correlated gap is indirect and negative, yielding a semimetal state (therefore Hartree-Fock theory may not be accurate in those regimes).

and

$$\Sigma_{\text{F},gg'}^{\sigma}(\mathbf{k}) = -\frac{1}{(2\pi)^2} \sum_{g''} \int_{\text{mBZ}} d^2\mathbf{q} C_{\mathbf{p}-\mathbf{k}+\mathbf{g}''} \rho_{(\mathbf{g}+\mathbf{g}'')(\mathbf{g}'+\mathbf{g}'')}^{\sigma}(\mathbf{q}), \quad (\text{S9})$$

where $C_{\mathbf{q}} = 2\pi e^2/\epsilon|\mathbf{q}|$ is the Fourier-transformed Coulomb potential with C_0 set to 0 by working relative to a uniform neutralizing background [94]. mBZ stands for the folded moiré BZ corresponding to the $\sqrt{2} \times \sqrt{2}$ supercell. Given filling factor ν^{σ} in each spin sector σ , *i.e.*, the number of doping electrons per $\sqrt{2} \times \sqrt{2}$ supercell in each spin sector, the density matrix is in turn determined by the mean-field Hamiltonian via

$$\rho^{\sigma}(\mathbf{k}) = \sum_{n=1}^{\nu^{\sigma}} |\phi_{n\mathbf{k}}^{\sigma}\rangle \langle \phi_{n\mathbf{k}}^{\sigma}|, \quad (\text{S10})$$

where $|\phi_{n\mathbf{k}}^{\sigma}\rangle$ is the n th normalized eigenvector of $H_{\text{MF}}^{\sigma}(\mathbf{k})$. Self-consistency in the relations between ρ^{σ} and H_{MF}^{σ} is reached via iteration assisted with the direct inversion iterative subspace method [95–97].

With the doping density corresponding to one electron per *original* moiré cell (*i.e.*, 2 electrons per supercell), we run the Hartree-Fock iterative algorithm under both spin-resolved doping configurations, $(\nu^{\uparrow}, \nu^{\downarrow}) = (1, 1)$ and $(2, 0)$, to address possible competition between antiferromagnetic and ferromagnetic couplings [70] in the electron spins. For each $(\nu^{\uparrow}, \nu^{\downarrow})$, we start from various independent random initial guesses of the density matrices. For the convergent solutions, we select the one with the lowest total mean-field energy

$$E_{\text{MF}} = \sum_{\mathbf{k} \in \text{mBZ}} \sum_{\sigma=\uparrow,\downarrow} \text{tr} \left[\rho^{\sigma}(\mathbf{k}) \left(H_0(\mathbf{k}) + \frac{\Sigma_{\text{H}} + \Sigma_{\text{F}}^{\sigma}(\mathbf{k})}{2} \right) \right]. \quad (\text{S11})$$

It turns out that under all parameter combinations involved in our Hartree-Fock calculations in this work, the spin-compensated solution $((\nu^{\uparrow}, \nu^{\downarrow}) = (1, 1))$ always has lower E_{MF} than spin-polarized solutions $((\nu^{\uparrow}, \nu^{\downarrow}) = (2, 0))$.

Fig. S2 presents several maps of band splitting $\Delta E_n = \max_{\mathbf{k}} |\Delta E_{n\mathbf{k}}|$ vs model parameters in our domain-centered system. We see that spin splitting is persistent across wide range in the parameter space. We also confirm that in

all the presented parameter regimes the splitting is d -wave by checking $\max_{\mathbf{k}} |\Delta E_{n\mathbf{k}} + \Delta E_{nC_4\mathbf{k}}|$ at each point, which turns out to always be negligible compared to the spin splitting. Here C_4 is counterclockwise rotation by 90° .

Scaling invariance

We consider two moiré lattice systems, both with relaxation governed by Eq. (S6) but one of them with primed notations:

$$\mu' \nabla^2 \mathbf{u}' + (\lambda' + \mu') \nabla (\nabla \cdot \mathbf{u}') + (\mu'_s - \mu') (\partial_y, \partial_x) (\partial_x u'_y + \partial_y u'_x) = i \sum_{\mathbf{G}} \mathbf{G} V'_{\mathbf{G}} e^{i\mathbf{G} \cdot (\mathcal{M}' \mathbf{r} + \mathbf{u}'(\mathbf{r}))}. \quad (\text{S12})$$

Let their parameters relate by $\lambda = \alpha_\mu \lambda$, $\mu' = \alpha_\mu \mu$, $\mu'_s = \alpha_\mu \mu_s$ and $\mathcal{M}' = \alpha_\epsilon \mathcal{M}$, and let their spatial configurations relate by $\mathbf{u}'(\alpha_r \mathbf{r}) = \alpha_a \mathbf{u}(\mathbf{r})$. Eq. (S12) now becomes

$$\mu \nabla^2 \mathbf{u} + (\lambda + \mu) \nabla (\nabla \cdot \mathbf{u}) + (\mu_s - \mu) (\partial_y, \partial_x) (\partial_x u_y + \partial_y u_x) = i \frac{\alpha_r^2}{\alpha_\mu \alpha_a} \sum_{\mathbf{G}} \mathbf{G} V'_{\mathbf{G}} e^{i\mathbf{G} \cdot (\alpha_r \alpha_\epsilon \mathcal{M} \mathbf{r} + \alpha_a \mathbf{u}(\mathbf{r}))}. \quad (\text{S13})$$

One can show that if $\alpha_r = \alpha_a / \alpha_\epsilon$ and $V'_{\mathbf{G}} = \alpha_\mu \alpha_\epsilon^2 V_{\alpha_a \mathbf{G}}$, then the equation reproduces Eq. (S6). Therefore, if ϵ changes, the relative relaxation configuration can be kept invariant by properly scaling other parameters.

For electrons, we use more straight-forward arguments – the physics is invariant when all moiré energy scales vary proportionally. Hence, the amplitude of deformation potential $D \nabla \cdot \mathbf{u}(\mathbf{r})$ should scale with the moiré kinetic energy $E_m = \epsilon^2 \hbar^2 / 2ma^2$; $\nabla \cdot \mathbf{u}$ scales with $a/a_M \propto \epsilon$, where a_M is the moiré period, hence D scales with ϵ/ma^2 . To make the moiré-scale Coulomb interaction energy $C_m = |\epsilon|e^2/\tilde{\epsilon}a$ also scale with E_m , the inverse dielectric constant $1/\tilde{\epsilon}$ should scale with $|\epsilon|/ma$. We note that the above scaling arguments for both lattice relaxation and electrons also hold even when $\alpha_\epsilon, \alpha_r < 0$ (in which case D changes sign).

We note a subtlety in our scaling argument – since θ , B_{1g} and B_{2g} need to scale together with ϵ , when either B_{1g} or B_{2g} is nonzero, scaling in B_{1g} and B_{2g} can slightly deviate the shape of the pre-relaxation strained elastic lattice and thus also the shape of the moiré periodicity. However, we expect the effects of this deviation on lattice relaxation and moiré bands to be small within the continuum regime where $|\epsilon|, |\theta|, |B_{1g}|, |B_{2g}| \ll 1$.